

BJT Logic Gates

ENGS152 Circuits Lab

18 April 2026

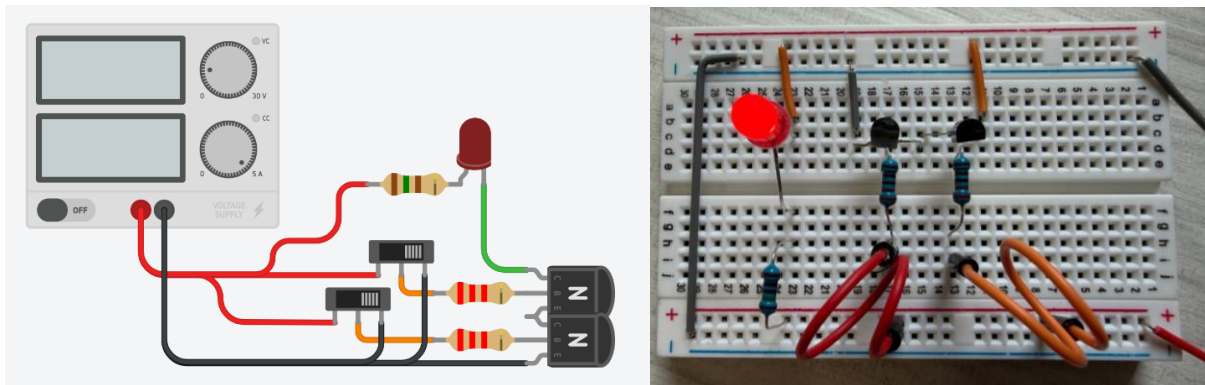
Step 1: AND Gate

We are given two inputs, **Input A** and **Input B**, the output Y behaves as shown in the truth table below.

Inputs		Outputs
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

The logic function represented by the truth table is a **2-input AND Gate**. It has an output of “1” if both of the inputs are “1”, otherwise it is “0”.

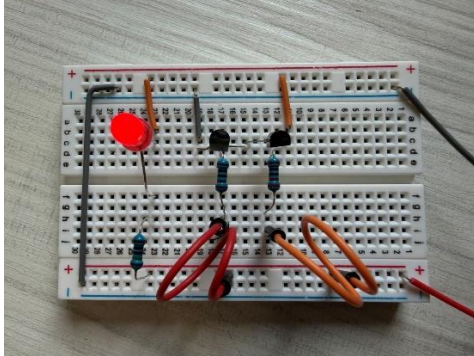
Below you can see the schematic circuit and real-life assembly on the breadboard.



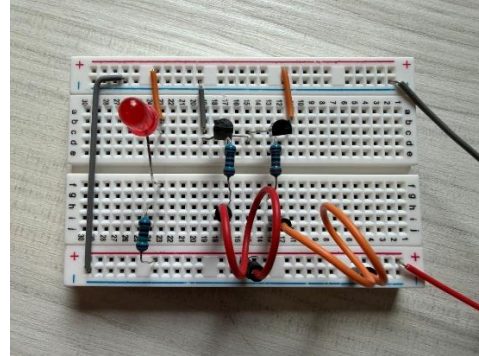
Note that jumper wires have been used as switches, to turn on or turn off the inputs, by connecting input either to 5V or GND

Below you can see the 2-input AND Gate in different states and see their corresponding outputs:

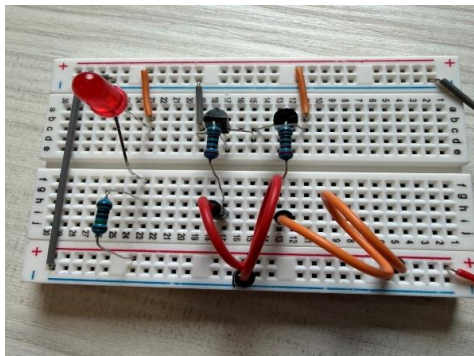
(A: 1, B: 1, Output: 1)



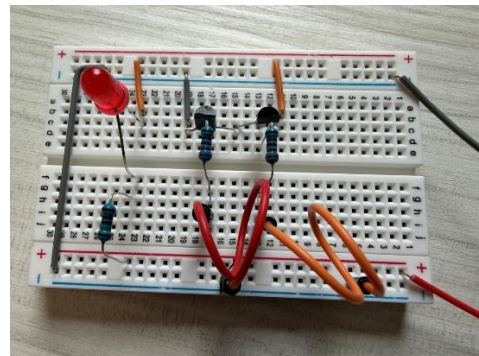
(A: 1, B: 0, Output: 0)



(A: 0, B: 1, Output: 0)



(A: 0, B: 0, Output: 0)



We have chosen this configuration because it is cost-efficient. This exact configuration opens the ground for the current flowing through the LED and it saves us a transistor compared to the configuration where a 3-input NAND Gate and an Inverter is used. The calculations are shown below.

$$V_L = 1.8 \text{ V}, I_C = 20 \text{ mA}, V_{CE1+CE2} = 0.4 \text{ V}, \beta = 10$$

$$I_B = 2 \text{ mA}$$

$$R_L = \frac{5 - 1.8 - 0.4 \text{ V}}{20 \text{ mA}} = 140 \Omega$$

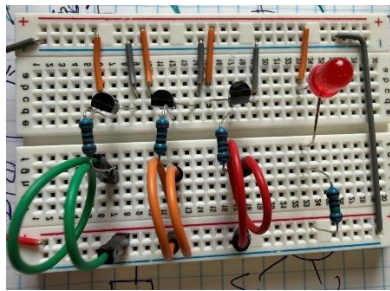
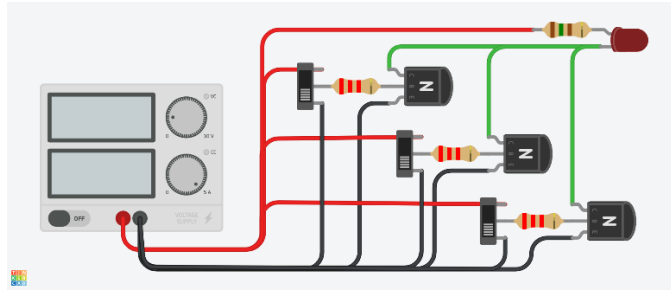
$$R_C = \frac{5 - 0.7 \text{ V}}{2 \text{ mA}} = 2150 \Omega$$

Take 150Ω as load resistor and $2.2 \text{ k}\Omega$ for base resistors.

Step 2: 3-input OR Gate

Now we are asked to design a circuit that implements the following Boolean function using basic BJT switching circuits:

$$Y = A + B + C$$



We have chosen this configuration because it is cost-efficient. This exact configuration opens the ground for the current flowing through the LED and it saves us a transistor compared to the configuration where a 3-input NOR Gate and an Inverter is used. We basically use 3 transistors instead of 4. The calculations are shown below.

$$V_L = 1.8 \text{ V}, I_C = 20 \text{ mA}, V_{CE} = 0.2 \text{ V}, \beta = 10, I_B = 2 \text{ mA}$$

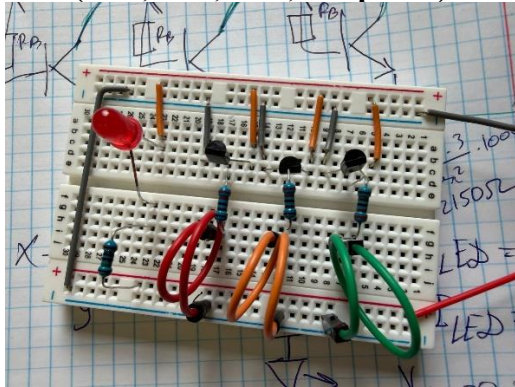
$$R_L = \frac{5 - 1.8 - 0.2 \text{ V}}{20 \text{ mA}} = 150 \, \Omega, R_C = \frac{5 - 0.7 \text{ V}}{2 \text{ mA}} = 2150 \, \Omega$$

Take $150 \, \Omega$ as load resistor and $2.2 \text{ k}\Omega$ for base resistors.

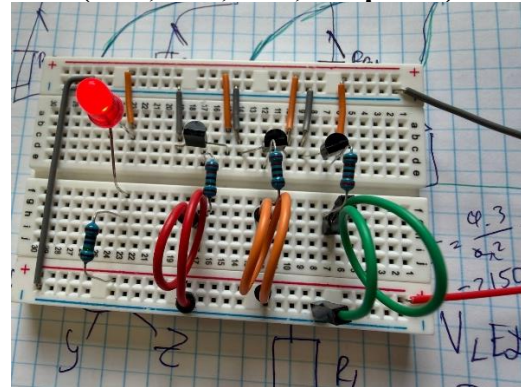
Input A	Input B	Input C	X = (A, B, C)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Let's look at some of the states and verify the truth table with our assembled circuit.

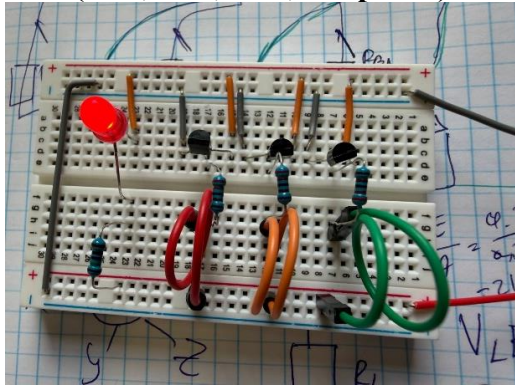
(A: 0, B: 0, C: 0, Output: 0)



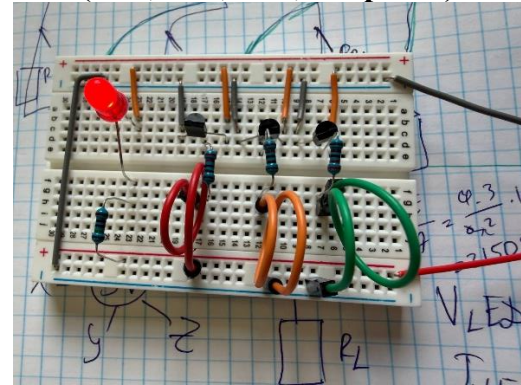
(A: 0, B: 1, C: 0, Output: 1)



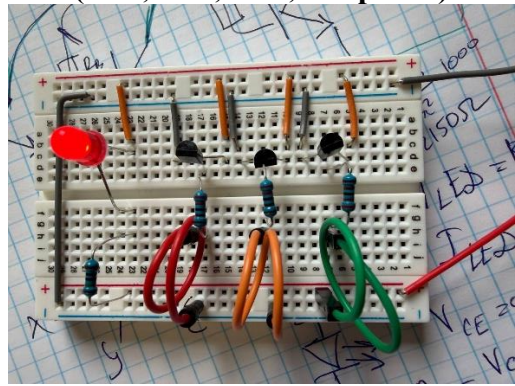
(A: 1, B: 1, C: 1, Output: 1)



(A: 1, B: 0, C: 0, Output: 1)



(A: 0, B: 0, C: 1, Output: 1)



We can see that even if one input is ON, the output also turns ON, this verifies the truth table.